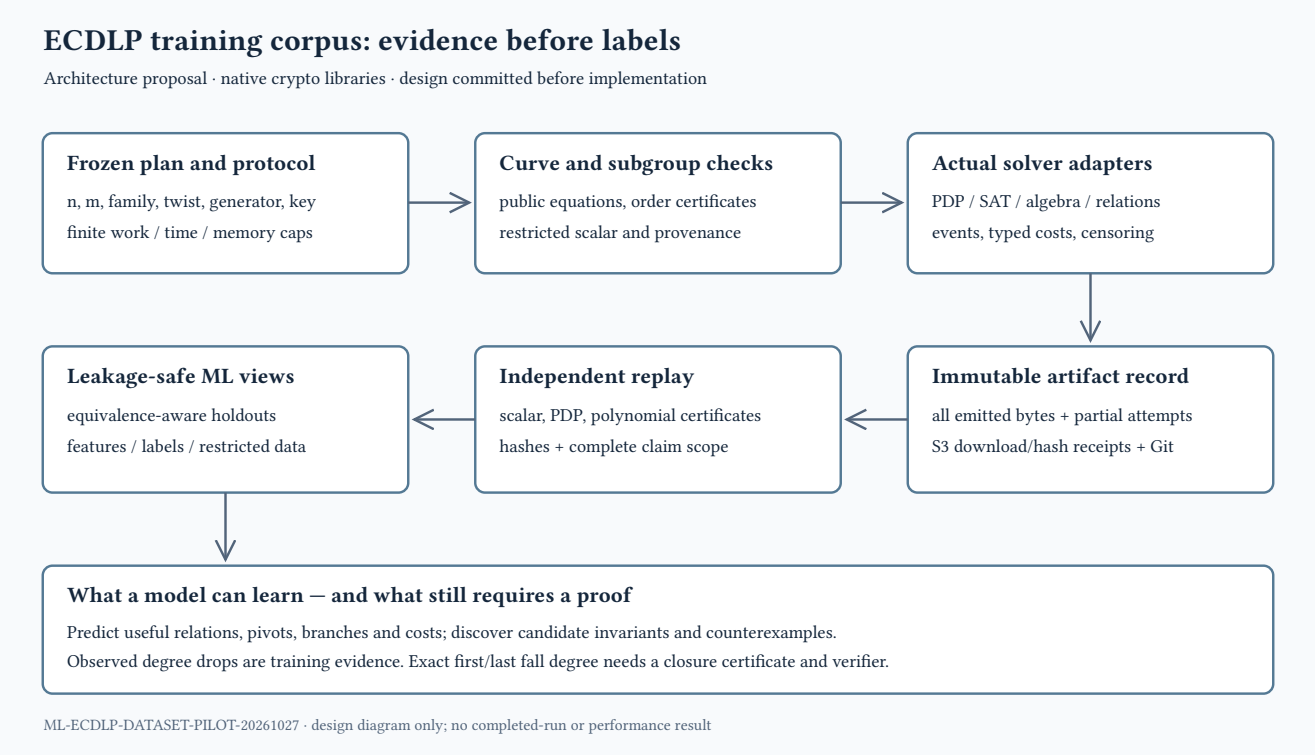
ECDLP corpus for machine-assisted algebra

Design and implementation scope, 27 October 2026 (client date).

This report describes a proposed dataset and its evidence contract. It reports no experiment result, speed comparison, solved campaign or mathematical verdict. The primary design is `docs/ml-dataset/DESIGN.md`, committed before implementation at `a5062b8ce4f`. The pilot identifier is `ML-ECDLP-DATASET-PILOT-20261027`.



The generator crosses Koblitz, ordinary binary and prime curves with paired twists, arities 2 through 15, field sizes 23 through 31 and 47 through 61, and multiple generators and target keys. At prime n=61 it uses the Mersenne prime 2305843009213693951. The default plan derives 165,150,720 method cells from

24 × 14 × 3 × 2 × 16 × 8 × 64 × 10. These are planned cells, not completed solutions. Capability and failure records must preserve every requested axis.

What the network can learn

Useful supervised targets include decomposition existence under an exact factor base; which candidate relation will add rank; which residual pair closes a large-prime cycle; which polynomial row or branch will simplify the current system; and which SAT decision or restart policy reduces independently counted remaining work. Sequence models can consume solver prefixes; graph models can consume variable, monomial, clause and relation incidence graphs.

Representation learning can test whether invariant features transfer across generators, twists, basis choices and field sizes. Predictions should be checked against sign and Frobenius actions explicitly. Runtime and solve-probability models must retain timeouts, unsupported cases and capped searches as censored outcomes. A selected successful subset would teach the wrong distribution.

Labels and mathematical boundaries

A constructed scalar d with Q = dG supplies a known answer. It does not show that an algorithm recovered d from G and Q. A verified PDP witness shows that a specified signed factor-base sum equals its target. It does not alone solve the ECDLP. An independent recovery label requires an algorithm that never reads the construction witness, plus separate scalar-multiplication replay.

First-fall and last-fall degrees are properties of stated polynomial systems and degree-closure spaces. A solver's maximum degree or first observed drop is a separate measured quantity. Models can suggest a low-degree identity, syzygy or counterexample. A checker must establish its algebraic hypotheses, membership and ranks before it becomes a proof or disproof. Finite examples do not establish a universal degree bound.

Preservation and cost measurement

Each cell binds its source commit, executable, protocol, plan, inputs, output artifacts and resource limits. S3 publication requires a byte-for-byte download and SHA-256 check, including the completion receipt. Interrupted attempts are preserved. Disjoint shards and external Job reconciliation prevent duplicate campaign ownership; local process locks alone cannot do that across hosts.

Typed native event counts, hardware instructions, cycles, CPU time, memory, artifact bytes and diagnostic wall time remain separate. Hardware counters retain raw and scaled readings, time enabled/running, inheritance and errors. Unavailable or interrupted readings remain null. Private native solver states that have no event hook are explicit trace gaps. Saving emitted artifacts is not a claim that every internal machine state has been observed.

Training split and validation

Keep scalars, seeds, future pivots, recovered logs and future learned clauses out of input features. Partition equivalent presentations, signs, Frobenius orbits, related generators and paired twists together. Start conservatively with field-size components, then merge components linked by embeddings or other cross-size constructions. Compare learned policies with frozen classical baselines, ablations and label-permutation controls under a separate approved training and evaluation protocol.

Sources and remaining work

The technical map is grounded in the repository's native Koblitz index-calculus, chained S3 SAT, matrix F4/F5, geometric F6, root, PKM and large-prime modules. Definitions and caveats for closure-based degree arguments were read at arXiv:1505.02532. PKM bibliographic context was checked against the University of Birmingham record for Algebraic approaches for the elliptic curve discrete logarithm problem. The full publisher paper was not retrieved.

The campaign must still establish verified native CLI behavior, bounded cloud execution, complete requested backend coverage, streaming trace completeness, consumer/export compatibility and durable generation waves. Those acceptance items remain distinct from this design record and from a successful pilot.